Nitroglycerin (glyceryl trinitrate): Drug information - UpToDate

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Contributor Disclosures

For additional information see ”Nitroglycerin (glyceryl trinitrate): Patient drug information” and ”Nitroglycerin (glyceryl trinitrate): Pediatric drug information”

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- GoNitro [DSC];
- Nitro-Bid;
- Nitro-Dur;
- Nitro-Time;
- Nitrolingual;
- NitroMist [DSC];
- Nitrostat;
- Rectiv

Brand Names: Canada

- MYLAN-Nitro;
- Nitro-Dur;
- Nitroject;
- Nitrolingual;
- RHO-Nitro;
- Trinipatch 0.2;
- Trinipatch 0.4;
- Trinipatch 0.6

Pharmacologic Category

- Antianginal Agent;
- Antidote, Extravasation;
- Vasodilator

Dosing: Adult

Dosage guidance:

Safety: If a patient who has taken a phosphodiesterase-5 inhibitor for erectile dysfunction develops chest pain, delay nitrate therapy for ≥12 hours after taking avanafil, ≥24 hours after taking sildenafil or vardenafil, and ≥48 hours after taking tadalafil (Ref).

Dosage form information: GoNitro sublingual powder has been discontinued in the United States for >1 year.

Acute decompensated heart failure

Acute decompensated heart failure (adjunctive agent):

Note: May consider in patients with volume overload in the absence of symptomatic hypotension to help relieve congestion and dyspnea as an adjunct to IV diuretics (Ref).

Continuous infusion: IV: Initial: 5 to 10 mcg/minute; titrate as needed based on response and tolerability in increments of 5 to 10 mcg/minute every 3 to 5 minutes up to 200 mcg/minute. Lower doses produce venous dilation; however, arterial vasodilation may occur at high doses. Tachyphylaxis develops within 24 to 48 hours of continuous administration; transition to long-term oral heart failure therapies as soon as possible for ongoing hemodynamic control (Ref).

Anal fissure

Anal fissure (alternative agent):

Note: Administer topically as a local vasodilator in conjunction with supportive measures. A 0.2% ointment is not commercially available and must be prepared by a licensed compounding facility (Ref).

Peri-anal 0.2% or 0.4% ointment: After cleansing, apply around fissure(s) twice daily as directed for 4 weeks; if symptoms persist, continue treatment for another 4 weeks for a total duration of 8 weeks (Ref).

Angina

Angina:

Note: Recommended for acute angina. For prevention of recurrent angina, may use in combination with other anti-anginal therapy (eg, a beta-blocker) (Ref).

Acute angina:

Note: If pain is not relieved or worsens 3 to 5 minutes after 1 sublingual or translingual dose, seek immediate emergency medical attention (eg, call 911) (Ref).

Sublingual powder (0.4 mg/packet): Initial: 1 packet at onset; repeat every 5 minutes if angina persists; may administer up to 3 packets in a 15-minute period (Ref).

Sublingual tablet: Initial: 0.3 or 0.4 mg at onset; repeat every 5 minutes if angina persists; may administer up to 3 tablets in a 15-minute period (Ref). For patients with refractory angina in the emergency department, up to 0.6 mg as a single dose may be considered (Ref).

Translingual 0.4 mg/spray: Initial: 1 spray at onset; repeat every 5 minutes if angina persists; may administer up to 3 sprays in a 15-minute period (Ref).

Continuous IV infusion:

Note: Consider IV therapy if angina is not relieved with other dosage forms. Avoid in patients with hypotension (eg, systolic BP <90 mm Hg or >30 mm Hg below baseline), marked bradycardia (eg, <50 beats per minute) or tachycardia (eg, >100 beats per minute), and/or suspected right ventricular infarction (Ref).

Initial: 5 to 10 mcg/minute with continuous cardiac monitoring; titrate as needed to relieve angina symptoms in increments of 5 mcg/minute every 5 to 10 minutes up to 20 mcg/minute; if angina persists at a dose of 20 mcg/minute, may increase by 10 to 20 mcg/minute every 3 to 5 minutes to a maximum dose of 400 mcg/minute (Ref). Tachyphylaxis develops within 24 to 48 hours of continuous nitrate administration.

Prevention of angina (adjunctive therapy):

Sublingual powder (0.4 mg/packet): Initial: 1 packet 5 to 10 minutes prior to activities that may provoke angina.

Sublingual tablet: Initial: 0.3 or 0.4 mg 5 to 10 minutes prior to activities that may provoke angina.

Translingual 0.4 mg/spray: Initial: 1 or 2 sprays 5 to 10 minutes prior to activities that may provoke angina.

Topical 2% ointment: Initial: Apply ½ inch upon rising and apply another ½ inch 6 hours later; if necessary, the dose may be doubled to 1 inch and subsequently doubled again to 2 inches if response is inadequate. Recommended maximum frequency of administration is 2 doses/day. Include a nitrate-free interval of ~10 to 12 hours each day to minimize the risk of tolerance.

Topical patch, transdermal: Initial: 0.2 to 0.4 mg/hour; increase dose as needed based on response and tolerability to a maximum of 0.8 mg/hour. Use a patch-on period of 12 to 14 hours/day and patch-off period (nitrate-free interval) of 10 to 12 hours/day to minimize the risk of tolerance.

ER capsule, oral: Initial: 2.5 to 6.5 mg 3 to 4 times daily; increase dose as needed based on response and tolerability to 26 mg 4 times daily. Include a nitrate-free interval of ~10 to 12 hours each day to minimize the risk of tolerance.

Extravasation management, sympathomimetic vasopressors

Extravasation management, sympathomimetic vasopressors (alternative agent) (off-label use):

Note: Nitroglycerin is an alternative if phentolamine is not available (Ref).

Topical 2% ointment: Apply a 1-inch strip to site of ischemia to cover the affected area; may repeat every 8 hours if needed (Ref).

Hypertension, perioperative

Hypertension, perioperative (alternative agent):

Note: For patients with chronic hypertension prior to surgery, restart oral therapies as soon as appropriate once hemodynamically stable. Address underlying causes (eg, pain, agitation, withdrawal, hypervolemia) prior to initiating antihypertensive therapy. Generally used for patients with hypervolemia who are not responsive to IV diuretic therapy, particularly if there is known or suspected ischemic heart disease or heart failure (Ref).

Continuous IV infusion: Initial: 5 to 10 mcg/minute; increase based on BP response and tolerability in increments of 5 mcg/minute every 3 to 5 minutes up to 20 mcg/minute; if no response at a dose of 20 mcg/minute, may increase by 10 to 20 mcg/minute every 3 to 5 minutes to a maximum dose of 200 mcg/minute. Lower doses primarily produce venous dilation; however, arterial vasodilation may occur at high doses (Ref). Tachyphylaxis develops within 24 to 48 hours of continuous nitrate administration; if vasodilator requirements continue longer than 24 to 48 hours, transition to an alternative IV or oral vasodilator.

Hypertensive emergency

Hypertensive emergency (alternative agent) (off-label use):

Note: Limitations of use include variable efficacy compared to other agents (eg, inconsistent and transient BP response), possible reflex tachycardia, and possible reduced cardiac output. May be used as adjunctive therapy for patients with acute coronary syndrome or acute pulmonary edema. When used for hypertensive emergency without a compelling condition, in general, reduce mean arterial pressure gradually by ~10% to 20% over the first hour. If stable, may then reduce to a goal BP of <160/100 mm Hg for the next 2 to 6 hours, then to 130 to 140 mm Hg over the next 24 to 48 hours to limit target organ injury. Invasive BP monitoring is recommended for appropriate dose titration (Ref).

Continuous IV infusion: Initial: 5 mcg/minute; increase based on BP response and tolerability in increments of 5 mcg/minute every 3 to 5 minutes up to 20 mcg/minute; if no response at a dose of 20 mcg/minute, may increase by 10 to 20 mcg/minute every 3 to 5 minutes to a maximum dose of 200 mcg/minute. Lower doses produce venous dilation; however, arterial vasodilation may occur at high doses (Ref). Tachyphylaxis develops within 24 to 48 hours of continuous nitrate administration; if vasodilator requirements continue longer than 24 to 48 hours, transition to an alternative IV or oral vasodilator.

Tendinopathy, persistent

Tendinopathy, persistent (adjunctive agent) (off-label use):

Note: May use in conjunction with physical rehabilitation (eg, resistance exercise program) or if other conservative measures (eg, activity modification, correction of biomechanical flaws, anti-inflammatory medications) do not adequately resolve symptoms (Ref).

Topical patch, transdermal: 0.05 to 0.1 mg/hour applied to the area of maximal tenderness; replace patch every 24 hours; may cut patch to size to administer the intended dose (eg, if desired dose is 0.05 mg/hour, then instruct patient to apply one-fourth (1/4) of a 0.2 mg/hour patch); it may take 3 to 6 months to experience significant improvement (Ref).

Uterine relaxation

Uterine relaxation (off-label use):

Note: May be used for obstetric emergencies (eg, uterine inversion, difficult fetal extraction due to uterine contraction, internal podalic version of a second twin) or to facilitate extraction of a trapped placenta, external cephalic version, or replacement of deeply prolapsed fetal membranes before placement of a cerclage (Ref). Dosing provided is an example and may vary based on indication.

IV: 50 mcg once; may repeat at 1-minute intervals as needed to sufficiently relax the uterus; maximum total dose: 250 mcg (Ref). If urgent uterine relaxation is required (eg, for fetal extraction), may use initial bolus of 100 to 200 mcg (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: IV, Oral, Peri-anal, Sublingual, Topical, Translingual, Transdermal: No dosage adjustment necessary for any degree of kidney dysfunction (<1% excreted in the urine) (Ref).

IV, Oral, Peri-anal, Sublingual, Topical, Translingual, Transdermal:

Hemodialysis, intermittent (thrice weekly): Unlikely to be dialyzed (large V_d): No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: Unlikely to be dialyzed (large V_d): No dosage adjustment necessary (Ref).

CRRT: No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer’s labeling.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Nitroglycerin (glyceryl trinitrate): Pediatric drug information")

Heart failure; cardiogenic shock

Heart failure; cardiogenic shock: Limited data available: **Note:** Continuous IV infusion dosing units vary by age (mcg/kg/minute or **mcg/minute**); extra precautions should be taken. In adults, tolerance to the hemodynamic and antianginal effects can develop within 24 hours of continuous use; nitrate-free interval (10 to 12 hours/day) is recommended to avoid tolerance development; gradually decrease dose in patients receiving nitroglycerin for prolonged periods to avoid withdrawal reaction (Ref).

Infants and Children: Continuous IV infusion: Initial: 0.25 to 0.5 mcg/kg/minute; titrate by 1 mcg/kg/minute every 15 to 20 minutes as needed; faster titration may be necessary in some patients; in adults, titration every 3 to 5 minutes has been suggested. Usual dose range: 1 to 5 mcg/kg/minute; usual maximum dose: 10 mcg/kg/minute (Ref); doses up to 20 mcg/kg/minute may be used (Ref).

Adolescents: Continuous IV infusion: Initial: 5 to 10 **mcg/minute**; titrate to a maximum dose of 200 **mcg/minute**. **Note:** In adults, doses are titrated every 3 to 5 minutes as needed (Ref).

Extravasation, treatment

Extravasation (sympathomimetic vasopressors), treatment (alternative to phentolamine): Very limited data available; dosing based on experience in neonatal patients; optimal dosing has not been established:

Infants, Children, and Adolescents: Topical: 2% ointment: 4 **mm**/kg applied as a thin ribbon to the affected areas; after 8 hours if no improvement, the dose may be repeated at the affected site (Ref). The maximum reported dose is application of a 1-inch strip to the affected site in a neonate (Ref); however, this is greater than the usual initial adult dose ($^1/_2$ inch) for angina; hypotension may occur; carefully monitor blood pressure (Ref). **Note:** Minimal data available from clinical trials/case reports; however, use has been described in reviews of extravasation treatment (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%: Nervous system: Headache (patch, rectal ointment: 63% to 64%; lingual spray, sublingual powder: >2%)

1% to 10%:

Cardiovascular: Hypotension (patch: ≤4%; including orthostatic hypotension), syncope (patch: ≤4%)

Nervous system: Dizziness (lingual spray, patch, rectal ointment, sublingual powder: 2% to 6%), paresthesia (lingual spray, sublingual powder: >2%)

Frequency not defined: Local: Application-site irritation (patch)

Postmarketing (any formulation):

Cardiovascular: Asystole (Thomas 2016), bradycardia (Brandes 1990), flushing, rebound hypertension, tachycardia, unstable angina pectoris, vasodilation (cutaneous)

Dermatologic: Contact dermatitis (Roche Gamón 2006), exfoliative dermatitis, fixed drug eruption, skin rash

Endocrine & metabolic: Lactic acidosis (Smith 2019)

Gastrointestinal: Nausea, vomiting

Hematologic & oncologic: Methemoglobinemia (Imani 2019)

Hypersensitivity: Hypersensitivity reaction (including nonimmune anaphylaxis)

Local: Application-site rash

Nervous system: Asthenia, drowsiness

Respiratory: Dyspnea, hypoxia (transient)

Contraindications

Hypersensitivity to nitroglycerin, other nitrates or nitrites, or any component of the formulation (includes adhesives for transdermal product); concurrent use with phosphodiesterase-5 (PDE-5) inhibitors (avanafil, sildenafil, tadalafil, or vardenafil); concurrent use with soluble guanylate cyclase (sGC) stimulators (eg, riociguat); acute circulatory failure or shock; early myocardial infarction (sublingual tablet only; see **Note** below); increased intracranial pressure; severe anemia.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Additional contraindications for IV product: Constrictive pericarditis; increased intracranial pressure; pericardial tamponade; restrictive cardiomyopathy.

Canadian labeling: Additional contraindications for IV product (not in manufacturer's US labeling): Hypotension; uncorrected hypovolemia.

Canadian labeling: Additional contraindications for translingual product (not in manufacturer's US labeling): Closed angle glaucoma; heart failure (aortic or mitral stenosis, constrictive pericarditis, or hypertrophic cardiomyopathy with left ventricular outflow tract obstruction).

Canadian labeling: Additional contraindications for transdermal patch (not in manufacturer's US labeling): orthostatic hypotension; myocardial insufficiency due to obstruction (eg, presence of aortic or mitral stenosis or of constrictive pericarditis); increased intraocular pressure.

Note : According to the American College of Cardiology/American Heart Association (ACC/AHA) guideline for the management of patients with acute coronary syndromes, avoid nitrates in the following conditions: Hypotension (SBP <90 mm Hg or >30 mm Hg below baseline), marked bradycardia or tachycardia, and right ventricular infarction (ACC/AHA [Rao 2025]).

Warnings/Precautions

Concerns related to adverse effects:

• Headache: Dose-related headaches may occur, especially during initial dosing.

• Hypotension/bradycardia: Severe hypotension and shock may occur (even with small doses); paradoxical bradycardia and increased angina pectoris may accompany hypotension. Orthostatic hypotension may also occur; ethanol may accentuate this. Use with caution in volume depletion, preexisting hypotension, constrictive pericarditis, aortic or mitral stenosis, and extreme caution with inferior wall myocardial infarction (MI) and suspected right ventricular involvement.

• Increased intracranial pressure: Nitroglycerin may precipitate or aggravate increased intracranial pressure and subsequently may worsen clinical outcomes in patients with neurologic injury (eg, intracranial hemorrhage, traumatic brain injury) (Rangel-Castilla 2008). Use is contraindicated in patients with increased intracranial pressure.

Disease-related concerns:

• Hypertrophic cardiomyopathy with left ventricular outflow tract obstruction: Avoid use in patients with hypertrophic cardiomyopathy with left ventricular outflow tract obstruction; nitrates may reduce preload, exacerbating obstruction and cause hypotension or syncope and/or worsening of heart failure (HF) (AHA/ACC [Ommen 2024]).

Dosage form specific issues:

• Intra-anal ointment: Use caution when treating rectal anal fissures with nitroglycerin in patients with suspected or known significant cardiovascular disorders (eg, cardiomyopathies, HF, acute MI); intra-anal nitroglycerin administration may decrease systolic BP and decrease arterial vascular resistance.

• Long-acting agents: Avoid use of long-acting agents in acute MI or acute HF; cannot easily reverse effects if adverse events develop.

• Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007).

• Transdermal patches: May contain conducting metal (eg, aluminum); remove patch prior to MRI.

Other warnings/precautions:

• Tolerance: May occur; cross tolerance to other nitro compounds have been reported. Appropriate dosing is needed to minimize tolerance development.

Warnings: Additional Pediatric Considerations

Some dosage forms may contain propylene glycol; in neonates, large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults, including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Product Availability

GoNitro sublingual powder has been discontinued in the United States for >1 year.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Aerosol Solution, Translingual:

NitroMist: 400 mcg/spray (4.1 g [DSC], 8.5 g [DSC]) [contains menthol]

Capsule Extended Release, Oral:

Nitro-Time: 2.5 mg [contains fd&c blue #1 (brilliant blue), fd&c red #40 (allura red ac dye), quinoline yellow (d&c yellow #10)]

Nitro-Time: 6.5 mg [contains fd&c blue #1 (brilliant blue), fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Nitro-Time: 9 mg [contains fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Ointment, Rectal:

Rectiv: 0.4% (30 g) [contains propylene glycol]

Generic: 0.4% (30 g)

Ointment, Transdermal:

Nitro-Bid: 2% (1 g, 30 g, 60 g)

Generic: 2% (1 g, 30 g)

Packet, Sublingual:

GoNitro: 400 mcg (1 ea [DSC])

Patch 24 Hour, Transdermal:

Nitro-Dur: 0.1 mg/hr (1 ea, 30 ea, 100 ea); 0.2 mg/hr (1 ea, 30 ea); 0.3 mg/hr (1 ea, 30 ea, 100 ea); 0.4 mg/hr (1 ea, 30 ea); 0.6 mg/hr (1 ea, 30 ea); 0.8 mg/hr (1 ea, 30 ea, 100 ea)

Generic: 0.1 mg/hr (1 ea, 30 ea); 0.2 mg/hr (1 ea, 30 ea); 0.4 mg/hr (1 ea, 30 ea); 0.6 mg/hr (1 ea, 30 ea)

Solution, Intravenous:

Generic: 25 mg (250 mL); 50 mg (250 mL); 100 mg (250 mL); 5 mg/mL (10 mL)

Solution, Translingual:

Nitrolingual: 0.4 mg/spray (4.9 g, 12 g)

Nitrolingual: 0.4 mg/spray (4.9 g [DSC], 12 g [DSC]) [contains alcohol, usp]

Generic: 0.4 mg/spray (4.9 g, 12 g)

Tablet Sublingual, Sublingual:

Nitrostat: 0.3 mg, 0.4 mg, 0.6 mg

Generic: 0.3 mg, 0.4 mg, 0.6 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Capsule Extended Release (Nitro-Time Oral)

2.5 mg (per each): $0.87

6.5 mg (per each): $1.00

9 mg (per each): $1.03

Ointment (Nitro-Biᑯ Transdermal)

2% (per gram): $1.56

Ointment (Nitroglycerin Rectal)

0.4% (per gram): $23.06

Ointment (Nitroglycerin Transdermal)

2% (per gram): $3.03

Ointment (Rectiv Rectal)

0.4% (per gram): $25.63

Patch, 24-hour (Nitro-Dur Transdermal)

0.1 mg/hour (per each): $28.12

0.2 mg/hour (per each): $30.25

0.3 mg/hour (per each): $63.59

0.4 mg/hour (per each): $33.91

0.6 mg/hour (per each): $36.78

0.8 mg/hour (per each): $68.96

Patch, 24-hour (Nitroglycerin Transdermal)

0.1 mg/hour (per each): $1.86 - $2.56

0.2 mg/hour (per each): $1.90 - $2.60

0.4 mg/hour (per each): $2.17 - $2.91

0.6 mg/hour (per each): $2.40 - $3.15

Solution (Nitroglycerin in D5W Intravenous)

100 mcg/mL 5% (per mL): $0.10

200 mcg/mL 5% (per mL): $0.11

Solution (Nitroglycerin Intravenous)

5 mg/mL (per mL): $2.03

Solution (Nitroglycerin Translingual)

0.4 mg/spray (per gram): $45.81

Solution (Nitrolingual Translingual)

0.4 mg/spray (per gram): $70.07

Sublingual (Nitroglycerin Sublingual)

0.3 mg (per each): $0.47

0.4 mg (per each): $1.02 - $1.03

0.6 mg (per each): $0.47

Sublingual (Nitrostat Sublingual)

0.3 mg (per each): $0.89

0.4 mg (per each): $1.94

0.6 mg (per each): $0.89

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Patch 24 Hour, Transdermal:

Nitro-Dur: 0.2 mg/hr (30 ea); 0.4 mg/hr (30 ea); 0.6 mg/hr (30 ea); 0.8 mg/hr (30 ea)

Trinipatch 0.2: 0.2 mg/hr (30 ea, 100 ea)

Trinipatch 0.4: 0.4 mg/hr (30 ea, 100 ea)

Trinipatch 0.6: 0.6 mg/hr (30 ea, 100 ea)

Generic: 0.2 mg/hr (30 ea); 0.4 mg/hr (30 ea); 0.6 mg/hr (30 ea); 0.8 mg/hr (30 ea)

Solution, Intravenous:

Nitroject: 5 mg/mL (10 mL) [contains alcohol, usp, propylene glycol]

Generic: 50 mg (250 mL); 5 mg/mL (10 mL); 400-5 MCG/ML-% (250 mL)

Solution, Translingual:

Nitrolingual: 0.4 mg/spray (4.9 g, 14.49 g) [contains alcohol, usp]

RHO-Nitro: 0.4 mg/spray (1 ea) [contains alcohol, usp]

Generic: 0.4 mg/spray ([DSC])

Administration: Adult

IV: Prepare in glass bottles or containers not made with PVC. Adsorption occurs to soft plastic (eg, PVC); use administration sets intended for nitroglycerin. Avoid in-line IV filters that adsorb nitroglycerin. Administer via infusion pump.

Intra-anal: Ointment: Using a finger covering (eg, plastic wrap, surgical glove, finger cot), place finger beside 1 inch measuring guide on the box and squeeze ointment the length of the measuring line directly onto covered finger. Insert ointment into the anal canal using the covered finger up to first finger joint (do not insert further than the first finger joint) and apply ointment around the side of the anal canal. If intra-anal application is too painful, may apply the ointment to the outside of the anus. Wash hands following application.

Oral:

ER capsule: Swallow whole. Do not chew, break, or crush. Administer with a full glass of water.

Bariatric surgery: Some institutions may have specific protocols that conflict with these recommendations; refer to institutional protocols as appropriate. Topical ointment and transdermal formulations are available. If safety and efficacy of nitroglycerin can be effectively monitored, no change in formulation or administration is required after bariatric surgery; however, selection of ointment, transdermal, or alternative therapy should be considered in high-risk patients.

Sublingual powder: Empty the contents of packet under the tongue, close mouth, and breathe normally. Allow powder to dissolve without swallowing. Do not rinse or spit for 5 minutes after dosing.

Sublingual tablet: Do not chew, crush, or swallow sublingual tablet. Place under tongue and allow to dissolve. Alternately, may be placed in the buccal pouch. May take small sip of water prior to placing tablet under the tongue to aid dissolution.

Translingual spray: Do not shake container. Prior to initial use, the pump must be primed by spraying 5 times (Nitrolingual) or 10 times (Nitromist) into the air. Priming sprays should be directed away from patient and others. Release spray onto or under tongue. Close mouth immediately after administration; do not inhale spray. Do not expectorate or rinse the mouth for 5 to 10 minutes following administration. Content of the container should be checked periodically; when the container is held upright, the end of the pump should be covered by the fluid in the bottle or the remaining sprays will not deliver the intended dose. If pump is unused for 6 weeks, a single priming spray (Nitrolingual) or 2 priming sprays (Nitromist) should be completed. If pump is unused for 3 months, re-prime with up to 5 sprays (Nitrolingual).

Topical:

Topical ointment: Wash hands prior to and after use. Application site should be clean, dry, and hair free. Use the provided paper applicator to measure the dose. Place paper on a flat surface, printed side down. Squeeze prescribed amount of ointment onto the paper. Place the paper (ointment side down) on the chest or back. Using the paper, spread in a thin layer over a 2.25 × 3.5 inch area. Do not rub into skin. Tape applicator into place.

Extravasation management, sympathomimetic vasopressors (alternative agent) (off-label use): Stop vesicant infusion immediately and disconnect IV line (leave needle/cannula in place); gently aspirate extravasated solution from the IV line (do NOT flush the line); remove needle/cannula; apply dry warm compresses; elevate extremity (Ref).

Alternatives to phentolamine:

Nitroglycerin topical 2% ointment: Apply a 1-inch strip to the site of ischemia to cover the affected areas; may repeat every 8 hours as necessary (Ref).

Topical patch, transdermal: Application site should be clean, dry, and hair free. Rotate patch sites. Dispose of any used or unused patches by folding adhesive ends together, replace in pouch or sealed container and discard properly in trash, away from children and pets.

Preparation for Administration: Adult

Vials: Dilute in a compatible solution (eg, D5W, NS); maximum concentration not to exceed 400 mcg/mL; prepare in glass bottles or containers not made with PVC (adsorption occurs to soft plastic [eg, PVC]); invert container several times to assure uniform dilution. Also available in premixed glass containers.

Usual Infusion Concentrations: Adult

Note: Premixed solutions available

IV infusion: 50 mg in 250 mL (concentration: 200 **mcg**/mL) **or** 100 mg in 250 mL (concentration: 400 **mcg**/mL) of D5W

Administration: Pediatric

Parenteral: Continuous IV infusion: Vials (concentrated solution): Not for direct injection; must be diluted prior to administration; premixed solution in glass containers (100 mcg/mL, 200 mcg/mL, 400 mcg/mL) is available. Administer via infusion pump. Adsorption to soft plastic (eg, PVC) occurs; special administration sets intended for nitroglycerin (nonpolyvinyl chloride) must be used; some inline IV filters also adsorb nitroglycerin; use of these filters should be avoided.

Topical ointment: Wash hands prior to and after use. Application site should be clean, dry, and hair free. Do not rub into skin.

Extravasation (sympathomimetic vasopressors), treatment: Stop vesicant infusion immediately and disconnect IV line (leave needle/cannula in place); gently aspirate extravasated solution from the IV line (do NOT flush the line); remove needle/cannula; elevate extremity. Apply nitroglycerin ointment as a thin ribbon to site of ischemia (Ref). May also apply dry warm compresses (Ref).

Preparation for Administration: Pediatric

Parenteral: Continuous IV infusion: Vial (concentrated solution): Dilute in D5W or NS to a maximum concentration of 400 mcg/mL; prepare in glass bottles, EXCEL or PAB containers (adsorption to soft plastic [eg, PVC] occurs). Premixed solution (100 mcg/mL, 200 mcg/mL, 400 mcg/mL) in glass containers is also available. ASHP recommends standard concentrations of 200 mcg/mL and 400 mcg/mL for pediatric patients (Ref).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions available.

IV infusion: 100 **mcg**/mL, 200 **mcg**/mL, or 400 **mcg**/mL.

Storage/Stability

ER capsules: Store at 25°C (77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from moisture.

IV:

Premixed (glass container): Store at 25°C (77°F); brief exposure up to 40°C (104°F) does not adversely affect the product. Protect from light until time of use. Avoid excessive heat; protect from freezing.

Vial: Store intact vial at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light until time of use. The manufacturer recommends only diluting in glass containers but does not provide storage condition recommendations (eg, temperature, duration) once diluted. Nitroglycerin diluted in D5W or NS to concentrations of 200 mcg/mL and 400 mcg/mL in glass or polyolefin containers were found to be stable for 28 days at room temperature or under refrigeration (Wagenknecht 1984). Also, refer to institution-specific policies and procedures.

Rectal ointment: Store at 20°C to 25°C (68°F to 77°F); excursions are permitted between 15°C and 30°C (59°F and 86°F). Keep the tube tightly closed. Use within 8 weeks of first opening.

Sublingual powder: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 5°C to 40°C (41°F to 104°F).

Sublingual tablets: Store at 20°C to 25°C (68°F to 77°F) in original glass container.

Transdermal ointment: Store at 20°C to 25°C (68°F to 77°F). Close tightly immediately after use.

Transdermal patch: Store at 15°C to 30°C (59°F to 86°F).

Translingual spray: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Do not forcefully open or burn container after use. Do not spray toward flames.

Use: Labeled Indications

Oral administration: Treatment or prevention of angina pectoris.

IV administration: Treatment of angina pectoris; acute decompensated heart failure; perioperative hypertension; induction of intraoperative hypotension.

Intra-anal administration (Rectiv ointment): Treatment of moderate to severe pain associated with chronic anal fissure.

Transdermal patch: Prevention of angina pectoris.

Use: Off-Label: Adult

Extravasation management, sympathomimetic vasopressors; Hypertensive emergency; Tendinopathy, persistent (adjunctive agent); Uterine relaxation

Medication Safety Issues

Sound-alike/look-alike issues:

Nitroglycerin may be confused with nitrofurantoin, nitroprusside

Nitro-Bid may be confused with Macrobid

Nitroderm may be confused with NicoDerm

Nitrol may be confused with Nizoral

Nitrostat may be confused with Nilstat, nystatin

Other safety concerns:

Nitro drip is an error-prone stemmed/coined drug name (mistaken as nitroprusside infusion)

Transdermal patch may contain conducting metal (eg, aluminum); remove patch prior to MRI.

International issues:

Nitrocor [Italy, Russia, and Venezuela] may be confused with Natrecor brand name for nesiritide [US, Canada, and multiple international markets]; Nutracort brand name for hydrocortisone in the [US and multiple international markets]; Nitro-Dur [US, Canada, and multiple international markets]

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Agents with Clinically Relevant Anticholinergic Effects: May decrease absorption of Nitroglycerin. Specifically, anticholinergic agents may decrease the dissolution of sublingual nitroglycerin tablets, possibly impairing or slowing nitroglycerin absorption. *Risk C: Monitor*

Alcohol (Ethyl): May increase vasodilatory effects of Vasodilators (Organic Nitrates). *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alteplase: Nitroglycerin may decrease serum concentrations of Alteplase. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Apomorphine: Nitroglycerin may increase hypotensive effects of Apomorphine. Management: Patients taking apomorphine should lie down before and after taking sublingual nitroglycerin. Monitor blood pressure for hypotension and orthostatic hypotension when these agents are combined. *Risk D: Consider Therapy Modification*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Dapoxetine: May increase orthostatic hypotensive effects of Vasodilators (Organic Nitrates). *Risk C: Monitor*

Dapsone (Topical): May increase adverse/toxic effects of Methemoglobinemia Associated Agents. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): May decrease vasodilatory effects of Nitroglycerin. Nitroglycerin may increase serum concentrations of Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates). Management: Avoid the use of ergot derivatives in patients receiving nitroglycerin for angina (or in any angina patient) if possible. If combined, monitor for decreased effects of nitroglycerin and increased adverse effects of the ergot derivative (eg, ergotism). *Risk D: Consider Therapy Modification*

Heparin: Nitroglycerin may decrease anticoagulant effects of Heparin. Nitroglycerin may decrease serum concentrations of Heparin. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Local Anesthetics: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Vasodilators (Organic Nitrates). *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitric Oxide: May increase adverse/toxic effects of Methemoglobinemia Associated Agents. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase vasodilatory effects of Vasodilators (Organic Nitrates). *Risk X: Avoid*

Prilocaine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Prilocaine. Combinations of these agents may increase the likelihood of significant methemoglobinemia. Management: Monitor for signs of methemoglobinemia when prilocaine is used in combination with other agents associated with development of methemoglobinemia. Avoid use of these agents with prilocaine/lidocaine cream in infants less than 12 months of age. *Risk C: Monitor*

Primaquine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Primaquine. Specifically, the risk for methemoglobinemia may be increased. Management: Avoid concomitant use of primaquine and other drugs that are associated with methemoglobinemia when possible. If combined, monitor methemoglobin levels closely. *Risk D: Consider Therapy Modification*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Rilmenidine: Vasodilators (Organic Nitrates) may increase hypotensive effects of Rilmenidine. *Risk C: Monitor*

Riociguat: Vasodilators (Organic Nitrates) may increase hypotensive effects of Riociguat. *Risk X: Avoid*

Rosiglitazone: Vasodilators (Organic Nitrates) may increase adverse/toxic effects of Rosiglitazone. Specifically, a greater risk of ischemia and other adverse effects has been associated with this combination in some pooled analyses. *Risk C: Monitor*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Sodium Nitrite: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Sodium Nitrite. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Tiapride: May increase hypotensive effects of Vasodilators (Organic Nitrates). *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Pregnancy Considerations

Nitroglycerin crosses the placenta (David 2000).

Following a single maternal IV bolus dose of nitroglycerin at the time of incision prior to cesarean delivery, concentrations in the umbilical cord at birth were significantly lower than the maternal plasma (~1 minute after dosing); a wide variation in maternal plasma concentrations was observed (David 2000). Following application of a transdermal patch 0.4 mg/hour to pregnant patients 20 to 36 weeks gestation, concentrations of nitroglycerin were low but detectable in the fetal serum ~1 to 4 hours after the patch was applied (fetal/maternal ratio: 0.23) (Bustard 2003).

IV nitroglycerin is recommended for use in pregnant patients with preeclampsia when severe hypertension is associated with pulmonary edema (ESC [Regitz-Zagrosek 2018]).

Rectal nitroglycerin may be considered for use in pregnant patients with anal fissures when conservative measures are not effective (Cross 2023).

Based on its ability to produce smooth muscle relaxation, nitroglycerin may be used in obstetrical procedures when immediate relaxation of the uterus is needed, such as: uterine inversion following delivery (ACOG 183 2017), uterine relaxation during removal of retained placental tissue (ASA 2016), and management of breech delivery (Caponas 2001; Cluver 2015). Additional data may be necessary to further define the role of nitroglycerin for preterm labor (ACOG 171 2016; Duckitt 2014, Zamani 2024).

Breastfeeding Considerations

It is not known if nitroglycerin is present in breast milk.

Data related to the use of nitroglycerin and breastfeeding are limited (Böttiger 2010; O'Sullivan 2011).

Adverse events were not noted in breastfeeding infants of mothers using topical nitroglycerin for anal fissures (Taylor 2008). Rectal nitroglycerin may be considered for use in lactating patients with anal fissures when conservative measures are not effective (Cross 2023).

Use of nitroglycerin ointment has been noted in a case report for the treatment of Raynaud phenomenon of the nipple. Breastfeeding was discontinued prior to therapy. Signs and symptoms resolved within a few weeks (O'Sullivan 2011).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother.

Monitoring Parameters

Blood pressure, heart rate; consult individual institutional policies and procedures

Mechanism of Action

Nitroglycerin forms free radical nitric oxide. In smooth muscle, nitric oxide activates guanylate cyclase which increases guanosine 3'5' monophosphate (cGMP) leading to dephosphorylation of myosin light chains and smooth muscle relaxation. Produces a vasodilator effect on the peripheral veins and arteries with more prominent effects on the veins. Primarily reduces cardiac oxygen demand by decreasing preload (left ventricular end-diastolic pressure); may modestly reduce afterload; dilates coronary arteries and improves collateral flow to ischemic regions. For use in rectal fissures, intra-anal administration results in decreased sphincter tone and intra-anal pressure.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Sublingual tablet: 1 to 3 minutes; Translingual spray: Similar to sublingual tablet; Extended release: ~60 minutes; Topical: 15 to 30 minutes; Transdermal: ~30 minutes; IV: Immediate

Peak effect: Sublingual powder: 7 minutes; Sublingual tablet: 5 minutes; Translingual spray: 4 to 15 minutes; Extended release: 2.5 to 4 hours; Topical: ~60 minutes; Transdermal: 120 minutes; IV: Immediate

Duration: Sublingual tablet: At least 25 minutes; Translingual spray: Similar to sublingual tablet; Extended release: 4 to 8 hours (Gibbons 2003); Topical: 7 hours; Transdermal: 10 to 12 hours; IV: 3 to 5 minutes

Distribution: V_d: ~3 L/kg

Protein binding: 60%

Metabolism: Extensive first-pass effect; metabolized hepatically to glycerol di- and mononitrate metabolites via liver reductase enzyme; subsequent metabolism to glycerol and organic nitrate; non-hepatic metabolism via red blood cells and vascular walls also occurs

Half-life elimination: ~1 to 4 minutes

Excretion: Urine (as inactive metabolites)

Patient Counseling Points

(For additional information see "Nitroglycerin (glyceryl trinitrate): Patient drug information")

What is this drug used for?

Rectal:

• It is used to treat anal pain.

Injection:

• It is used to treat high blood pressure.

• It is used to treat heart failure (weak heart).

• It is used to treat chest pain or pressure.

All other products:

• It is used to treat chest pain or pressure.

• It is used to prevent chest pain or pressure.

All products:

• It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Headache

• Burning or tingling of mouth

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Severe dizziness

• Passing out

• Persistent headache

• Fast heartbeat

• Slow heartbeat

• Flushing

• Blurred vision

• Dry mouth

• Sweating a lot

• Pale skin

• Severe nausea

• Severe vomiting

• Abnormal heartbeat

• Agitation

• Severe loss of strength and energy

• Chest pain

• Severe skin irritation

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Deponit | Nitro Dur | Nitroderm | Nitrol;
- (AR) Argentina: Dauxona | Enetege | Enetege-safejet | Minitran | Niglinar | Nitradisc | Nitro Dur | Nitroderm | Nitrogray | Oxycardin | Trinitron;
- (AT) Austria: Deponit | Nitro | Nitro Mack | Nitroderm | Nitrodur | Nitrolingual | Perlinganit | Rectogesic;
- (AU) Australia: Glyceryl trinitrat | Glyceryl trinitrate medsurge | Gtn pohl | Lycinate | Minitran | Nitradisc | Nitro bid | Nitro Dur | Nitrostat | Rectogesic | Transiderm Nitro;
- (BD) Bangladesh: Angicard | Nitrin | Nitrocap | Nitrovas | Rectocare | Repane | Trocer;
- (BE) Belgium: Deponit | Diafusor | Minitran | Nitro m bid | Nitroderm | Nitrodyl | Nitrolingual | Nysconitrine | Rectogesic | Trinipatch | Willlong;
- (BF) Burkina Faso: Myonit insta;
- (BG) Bulgaria: Nitroderm | Nitroglycerin | Nitronal Aqueous | Perlinganit;
- (BR) Brazil: Nitradisc | Nitroderm | Nitronal | Tridil;
- (CH) Switzerland: Deponit | Deponit 10 | Deponit 5 | Minitran | Minitran 10 | Nitro dur 10 | Nitro Mack | Nitroderm | Nitroglycerin Bioren | Nitroglycerin sintetica | Nitroglycerin wander | Nitrolingual | Perlinganit | Rectogesic | Trinitrosan;
- (CI) Côte d'Ivoire: Myonit insta;
- (CL) Chile: Angiolingual | Nitrocor | Nitroderm | Nitroglicerina;
- (CN) China: Deponit | Nitro Mack | Nitroderm | Nitrodisc | Nitroglycerin | Nitroglycerin patches (ii) | Nitrostat | Pai luo;
- (CO) Colombia: Enetege | Imectin | Nitradisc | Nitroglicerina | Triԁil;
- (CR) Costa Rica: Nitroglicerina;
- (CZ) Czech Republic: Deponit | Nitro | Nitroderm | Nitroglycerin | Nitromack | Nitropelet | Perlinganit;
- (DE) Germany: Aquo-trinitrosan | Corangin nitrospray | Coro nitro | Deponit | Glyceroltrinitrat carinopharm | Minitrans | Neos nitro opt | Nitradisc | Nitriderm | Nitro | Nitro Dur | Nitro gesanit | Nitro Mack | Nitro solvay infus n | Nitroderm | Nitrolingual | Nitronal | Perlinganit | Rectogesic | Trinitrosan;
- (DK) Denmark: Glycerylnitrat sad;
- (DO) Dominican Republic: Nitroderm;
- (EC) Ecuador: Nitroderm | Nitroglicerina | Nitroglicerina sanderson;
- (EE) Estonia: Deponit | Maycor nitrospray | Minitran | Myonit insta | Nitro | Nitro Mack | Nitroderm | Nitroglycerin | Nitroglycerin nycomed | Nitroglytseriin ns | Rectogesic;
- (EG) Egypt: Cardiocare | Coronolax | Deponit nt | Glycerotone | Nitro Mak Retard | Nitrocare | Nitrocine | Nitroderm | Nitromack | Nitronal | Nitrong | Nitroprinz | Nitrotard | Triԁil;
- (ES) Spain: Cordiplast | Dermatrans | Diafusor | Epinitril | Minitran | Nitradisc | Nitro Dur | Nitroderm | Nitroderm matrix | Nitrofix | Nitroplast | Rectogesic | Solinitrina | Trinipatch | Vernies;
- (FI) Finland: Deponit nt | Minitran | Nitro | Nitroglycerin orifarm | Nitroglycerin Orion | Nitromex | Nitropront | Nitroretard | Perlinganit | Rectogesic | Transiderm Nitro;
- (FR) France: Cordipatch | Diafusor | Discotrine | Epinitril | Lenitral percutane | Nitriderm | Nitronal | Rectogesic | Trinipatch | Trinitran | Trinitrine | Trinitrine laleuf | Trinitrine mylan | Trinitrine simple laleuf;
- (GB) United Kingdom: Deponit | Glyceryl trin dc | Glyceryl trin kent | Glyceryl trinitrate cox | Minitran | Minitran old | Nitro Dur | Nitrocine | Nitroglycerine | Nitronal | Percutol | Percutol rorer | Rectogesic | Retrogesic | Transiderm Nitro | Triԁil ahs | Trintek;
- (GR) Greece: Cardiplast | Carina | Epinitril | Nitro Mack | Nitrocerin | Nitrodyl | Nitroglicerina/acarpia | Nitrolingual | Nitrong | Nitroretard | Nitrosylon | Pancoran | Rectogesic | Sodemethin | Supranitrin | Trinipatch tts | Trinitrina;
- (GT) Guatemala: Nitroderm tts | Nitrofix | Nitroglicerina;
- (HK) Hong Kong: Angised | Deponit | Glyceryl trinitrat | Nitradisc | Nitro | Nitro bid | Nitro Mack | Nitroderm | Nitroglycerin | Nitroglyn | Nitrolingual;
- (HU) Hungary: Deponit | Gilustenon | Nitradisc | Nitro | Nitro Dur | Nitro Mack | Nitroderm | Nitroglicerina bioindustria | Nitrolingual | Nitromint | Nitromint nt | Perlinganit | Rectogesic;

- (ID) Indonesia: Carnit | Deponit | Glyceryl trinitrat | Minitran | Nitradisc | Nitro Mack | Nitrocine | Nitroderm | Nitroglycerin | Nitrokaf;
- (IE) Ireland: Angised | Deponit | Nitrocine | Nitrodur | Nitronal | Percutol | Rectogesic | Transiderm Nitro;
- (IL) Israel: Deponit | Nitroderm | Nitrovis;
- (IN) India: Angiplat | Angispan tr | Angistat | GTN Sorbitrate | Minitran | Myonit insta | Myovin | NG care | Nglong | Nitrocure | Nitroday | Nitroderm | Nitrogesic | Nitroglyn | Nitroject | Nitroplus | Nitrovir | Sorbitrate insta | Top nitro | Vasovin XL;
- (IT) Italy: Adesicor | Adesitrin | Deponit | Dermatrans | Epinitril | Keritrina | Minitran | Niglina | Niredil | Nitraket | Nitro Dur | Nitrocor | Nitroderm | Nitroglicerina bioindustria | Nitroglicerina doc generici | Nitroglicerina eg | Nitroglicerina pht | Nitroglicerina provvisoria | Nitroglicerina teva | Nitroglicerina zentiva | Nitrosylon | Perganit | Rectogesic | Top nitro | Triniplas | Trinitrina | Venitrin T;
- (JO) Jordan: Deponit | Nitrocine | Nitroderm;
- (JP) Japan: Dydrene | Herzer | Meditrans | Millis | Millisrol | Minitro | Myocor | Ngo | Nitroderm tts | Nitroglycerin | Nitroglycerin acc | Nitroglycerin hk | Nitroglycerin te | Nitroglycerin towa | Nitropen | Vasolator;
- (KE) Kenya: Niredil;
- (KR) Korea, Republic of: Angiderm | Deponit | Hana nitroglycerin | Millisrol | Minitran | Myungmoon nitroglycerin | Nitcerin | Nitro | Nitrobid | Nitroderm | Nitrogesic | Nitroglycerin | Nitrolingual | Nitrostat | Pasarect | Perlinganit | Rectocare | Rectogesic;
- (KW) Kuwait: Angised | Nitroderm | Tridil;
- (LB) Lebanon: Deponit | Discotrine | Lenitral | Nitro bid | Nitroderm | Nitromack | Rectogesic;
- (LT) Lithuania: Deponit | Deponit nt | Nitro | Nitro pohl | Nitrocine | Nitroderm | Nitroglycerin | Nitroglycerinum lek | Nitrogranulong | Nitromack | Perlinganit | Rectogesic | Trinitrolong;
- (LU) Luxembourg: Deponit | Diafusor | Minitran | Nitro m bid | Nitroderm | Nitrodyl | Nitrolingual | Nitromack | Nysconitrine | Rectogesic | Trinipatch | Willlong;
- (LV) Latvia: Minitran | Nitro | Nitro Dur | Nitrocine | Nitroderm | Nitroglycerin | Nitroglycerin dak | Nitroglycerinum | Nitrogranulong | Nitromack | Perlinganit | Rectogesic | Trinitrolong;
- (MA) Morocco: Lenitral | Nitroderm | Trinipatch;
- (MX) Mexico: Angiopohl | Brudanet | Cardinit | Minitran | Nitradisc | Nitro Dur | Nitroderm | Nitroderm tts;
- (MY) Malaysia: Angised | Deponit | Glyceryl trinitrat | Myonit insta | Nitrocine | Nitroderm | Nitroglycerin;
- (NL) Netherlands: Deponit | Minitran | Nitro Dur | Nitrobaat | Nitroglycerin | Nitroglycerine | Nitroglycerine teva | Nitrostat | Nitrozell | Transiderm | Transiderm Nitro | Trinipatch;
- (NO) Norway: Glyceroltrinitrat carinopharm | Glycerylnitrat | Glycerylnitrat i essex | Minitran | Nitradisc | Nitro Dur | Nitroglycerin | Nitroglycerin abboxia | Nitroglycerin macure | Nitroglycerin nycomed | Nitroglycerin orifarm | Nitromex | Nitronal | Nitroven | Rectogesic | Suscard | Transiderm Nitro;
- (NZ) New Zealand: Glyceryl trinitrat | Minitran | Nitroderm | Nitroglycerin | Nitronal | Nitronal Aqueous | Rectogesic;
- (PA) Panama: Minitran | Nitroglicerina;
- (PE) Peru: Deponit | Nitroglicerina | Nitropack d-r;
- (PH) Philippines: Deponit | Minitran | Nitronal Aqueous | Nitrosan | Nitrostat | Perlinganit | Transderm Nitro;
- (PK) Pakistan: Angilingual | Anginor | Angiocard | Angiocard sr | Angised | Deponit | Niglys | Nitromack | Nitronal | Nitroscot;
- (PL) Poland: Deponit | Lenitral | Nirmin | Nitracor | Nitro bid | Nitrocard | Nitroderm | Nitrodisk | Nitroglycerin | Nitroglycerinum | Nitromack | Perlinganit | Polnitrin | Rectogesic | Trinity;
- (PR) Puerto Rico: Deponit | Gonitro | Minitran | Nitrek | Nitro bid | Nitro Dur | Nitro Time | Nitrodil | Nitrodisc | Nitroglycerin | Nitroglyn | Nitrol | Nitronal | Nitroquick | Nitrostat | Nitrotab | Rectiv | Transderm Nitro | Tridil;
- (PT) Portugal: Epinitril | Glyceryl trinitrat | Gtn | Nitradisc | Nitro Dur | Nitroderm | Nitroglicerina | Nitroglicerina Abbott | Nitroglicerina mylan | Plastranit | Rectogesic | Trinipatch;
- (PY) Paraguay: Dauxona | Nitroglicerina sanderson;

- (QA) Qatar: Anrecta | Glytrin | Glytrin Spray | Myonit Insta | Nitro-BID | Nitrocin | Nitrocine | Nitroderm TTS 10 | Nitroderm TTS 15 | Nitroderm TTS 5 | Nitrolingual Spray | Nitronal | Nitrostat | Rectoderm | Rectogesic | Solintrina;
- (RO) Romania: Nitroretard;
- (RU) Russian Federation: Deponit | Nirmin | Nitro | Nitro Mack | Nitro Pohl Infuse | Nitro Time | Nitrocardin | Nitrocor | Nitroderm | Nitroglycerin | Nitromack | Nitropercuten | Nitrospray | Perlinganit | Trinitrolong;
- (SA) Saudi Arabia: Glyceryl trinitrat | Nitro Dur | Nitro Mack | Nitroderm | Tridil;
- (SE) Sweden: Minitran | Nitroglycerin | Nitroglycerin abboxia | Nitroglycerin biophausia | Nitroglycerin dak | Nitroglycerin Ebb | Nitroglycerin macure | Nitroglycerin Recip | Nitroretard | Rectogesic | Transiderm Nitro;
- (SG) Singapore: Angised | Deponit | Minitran | Nitrocine | Nitroderm;
- (SI) Slovenia: Minitran | Nitro Dur | Nitroderm | Nitrofix | Nitroglicerina sandoz | Nitronal;
- (SK) Slovakia: Nitroderm | Nitromack | Nitropelet | Perlinganit;
- (SR) Suriname: Myonit insta | Nitroderm tts | Nitroglycerine;
- (TH) Thailand: Deponit | Glyceryl trinitrate dbl | Minitran | Nitradisc | Nitro Mack | Nitrocine | Nitroderm | Nitroglycerin | Nitroject | Vasonit | Willong;
- (TN) Tunisia: Diafusor;
- (TR) Turkey: Anrecta | Deponit | Glisenit | Glyceryl trinitrat | Nitradisc | Nitroderm | Perlinganit | Rectoderm | Rectogesic | Trinity;
- (TW) Taiwan: Glyceryl trinitrat | Goyceryl trinitrat | Millisrol | Minitran | Nitro Mack | Nitroderm | Nitrodisc | Nitroglycerin | Nitrol | Nitrolingual | Nitrostat | Tridil;
- (UA) Ukraine: Deponit | Nitro | Nitro Mack | Nitro mick | Nitro Time | Nitroglycerin | Nitromack | Nitromax | Perlinganit | Trinitrolong;
- (UG) Uganda: Gtn 2.6 sr | Nitrocontin;
- (UY) Uruguay: Dauxona | Enetege | Minitran | Niglinar | Nitroderm | Nitroderm tts | Nitrodur | Nitroglicerina | Nitroglicerina d.n.s.ff.aa. | Nitroglicerina sanderson | Nitroglycerin;
- (VE) Venezuela, Bolivarian Republic of: Deponit | Minitran | Nitrocor | Nitroderm | Nitroglicerina | Nitromack;
- (VN) Viet Nam: Nicerol | Niglyvid | Nitralmyl;
- (ZA) South Africa: Nitrocine | Tridil

REFERENCES

1. Altabef KM, Spencer JT, Zinberg S. Intravenous nitroglycerin for uterine relaxation of an inverted uterus. *Am J Obstet Gynecol*. 1992;166(4):1237-1238. [PubMed 1566776]

2. American College of Obstetricians and Gynecologists (ACOG) Committee on Practice Bulletins—Obstetrics. ACOG Practice Bulletin No. 171: Management of preterm labor. *Obstet Gynecol*. 2016;128(4):e155-e164. doi:10.1097/AOG.0000000000001711 [PubMed 27661654]

3. American College of Obstetricians and Gynecologists (ACOG) Committee on Practice Bulletins—Obstetrics. ACOG Practice Bulletin No. 183: Postpartum hemorrhage. *Obstet Gynecol*. 2017;130(4):e168-e186. doi:10.1097/AOG.0000000000002351 [PubMed 28937571]

4. American Heart Association. Nitroglycerin. In: AHA 2016 PALS Digital Pocket Reference Card. Published 2016.

5. American Society of Anesthesiologists Task Force on Obstetric Anesthesia; Society for Obstetric Anesthesia and Perinatology. Practice guidelines for obstetric anesthesia: an updated report by the American Society of Anesthesiologists (ASA) Task Force on Obstetric Anesthesia and the Society for Obstetric Anesthesia and Perinatology. *Anesthesiology*. 2016;124(2):270-300. doi:10.1097/ALN.0000000000000935 [PubMed 26580836]

6. American Society of Health-System Pharmacists (ASHP). Pediatric continuous infusion standards. https://www.ashp.org/-/media/assets/pharmacy-practice/s4s/docs/Pediatric-Infusion-

Standards.ashx. Updated January 2021. Accessed February 9, 2022.

7. Artman M, Graham TP Jr. Guidelines for vasodilator therapy of congestive heart failure in infants and children. *Am Heart J*. 1987;113(4):994-1005. [PubMed 3031972]

8. Axemo P, Fu X, Lindberg B, Ulmsten U, Wessén A. Intravenous nitroglycerin for rapid uterine relaxation. *Acta Obstet Gynecol Scand*. 1998;77(1):50-53. doi:10.1034/j.1600-0412.1998.770111.x [PubMed 9492718]

9. Bacher H, Mischinger HJ, Werkgartner G, et al. Local nitroglycerin for treatment of anal fissures: an alternative to lateral sphincterotomy? *Dis Colon Rectum*. 1997;40(7):840-845. doi:10.1007/BF02055444 [PubMed 9221864]

10. Bayhi DA, Sherwood CD, Campbell CE. Intravenous nitroglycerin for uterine inversion. *J Clin Anesth*. 1992;4(6):487-488. doi:10.1016/0952-8180(92)90225-p [PubMed 1457119]

11. Bleday R. Anal fissure: medical management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed July 7, 2020.

12. Böttiger Y. Lack of Data on Glyceryl Trinitrate Risks in Breast Feeding. *Lakartidningen*. 2010;107(22):1483. [PubMed 20645601]

13. Brandes W, Santiago T, Limacher M. Nitroglycerin-induced hypotension, bradycardia, and asystole: report of a case and review of the literature. *Clin Cardiol*. 1990;13(10):741-744. doi:10.1002/clc.4960131015 [PubMed 2124178]

14. Broussard D, Ural K. Cardiovascular problems in the post-anesthesia care unit (PACU). Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed January 24, 2022.

15. Bustard MA, Ryan G, Seaward G, et al. Human Maternal and Fetal Plasma Glyceryl Trinitrate Concentrations. *Am J Obstet Gynecol*. 2003;189(6):1777-1778. [PubMed 14710114]

16. Caponas G. Glyceryl trinitrate and acute uterine relaxation: a literature review. *Anaesth Intensive Care*. 2001;29(2):163-177. [PubMed 11314836]

17. Challoumas D, Kirwan PD, Borysov D, Clifford C, McLean M, Millar NL. Topical glyceryl trinitrate for the treatment of tendinopathies: a systematic review. *Br J Sports Med*. 2019;53(4):251-262. doi:10.1136/bjsports-2018-099552 [PubMed 30301735]

18. Chedraui PA, Insuasti DF. Intravenous nitroglycerin in the management of retained placenta. *Gynecol Obstet Invest*. 2003;56(2):61-64. doi:10.1159/000072734 [PubMed 12900527]

19. Cluver C, Gyte GM, Sinclair M, Dowswell T, Hofmeyr GJ. Interventions for helping to turn term breech babies to head first presentation when using external cephalic version. *Cochrane Database Syst Rev*. 2015;(2):CD000184. doi:10.1002/14651858.CD000184.pub4 [PubMed 25674710]

20. Coons JC, McGraw M, Murali S. Pharmacotherapy for acute heart failure syndromes. *Am J Health Syst Pharm*. 2011;68(1):21-35. doi:10.2146/ajhp100202 [PubMed 21164062]

21. Cross KLR, Brown SR, Kleijnen J, et al. The Association of Coloproctology of Great Britain and Ireland guideline on the management of anal fissure. *Colorectal Dis*. 2023;25(12):2423-2457. doi:10.1111/codi.16762 [PubMed 37926920]

22. David M, Walka MM, Schmid B, et al. Nitroglycerin Application During Cesarean Delivery: Plasma Levels, Fetal/Maternal Ratio of Nitroglycerin, and Effects in Newborns. *Am J Obstet Gynecol*. 2000;182(4):955-961. [PubMed 10764480]

23. Dayan SS, Schwalbe SS. The use of small-dose intravenous nitroglycerin in a case of uterine inversion. *Anesth Analg*. 1996;82(5):1091-1093. doi:10.1097/00000539-199605000-00041 [PubMed 8610877]

24. Denkler KA, Cohen BE. Reversal of Dopamine Extravasation Injury With Topical Nitroglycerin Ointment. *Plast Reconstr Surg*. 1989;84(5):811-813. [PubMed 2510208]

25. Duckitt K, Thornton S, O'Donovan OP, Dowswell T. Nitric oxide donors for treating preterm labour. *Cochrane Database Syst Rev*. 2014;(5):CD002860. doi:10.1002/14651858.CD002860.pub2 [PubMed 24809331]

26. Dufour P, Vinatier D, Puech F. The use of intravenous nitroglycerin for cervico-uterine relaxation: a review of the literature. *Arch Gynecol Obstet*. 1997;261(1):1-7. doi:10.1007/s004040050189 [PubMed 9451516]

27. Elkayam U, Janmohamed M, Habib M, Hatamizadeh P. Vasodilators in the management of acute heart failure. *Crit Care Med*. 2008;36(1)(suppl):S95-S105. doi:10.1097/01.CCM.0000297161.41559.93 [PubMed 18158484]

28. Elliott WJ, Varon J. Drugs used for the treatment of hypertensive emergencies. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed January 24, 2022b.

29. Elliott WJ, Varon J. Evaluation and treatment of hypertensive emergencies in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed October 7, 2021a.

30. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

31. Friedman WF, George BL. Treatment of Congestive Heart Failure by Altering Loading Conditions of the Heart. *J Pediatr*. 1985;106(5):697-706. [PubMed 3923176]

32. Gibbons RJ, Abrams J, Chatterjee K, et al; American College of Cardiology; American Heart Association Task Force on practice guidelines (Committee on the Management of Patients With Chronic Stable Angina). ACC/AHA 2002 guideline update for the management of patients with chronic stable angina--summary article: a report of the American College of Cardiology/American Heart Association Task Force on practice guidelines (Committee on the Management of Patients With Chronic Stable Angina). *J Am Coll Cardiol*. 2003;41(1):159-168. [PubMed 12570960]

33. GoNitro (nitroglycerin) [prescribing information]. Canton, MS: Allegis Pharmaceuticals LLC; July 2019.

34. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on clinical practice guidelines. *Circulation*. 2022;145(18):e895-e1032. doi:10.1161/CIR.0000000000001063 [PubMed 35363499]

35. Hofmeyr GJ. Delivery of the singleton fetus in breech presentation. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed July 7, 2020.

36. Hollenberg SM. Vasodilators in acute heart failure. *Heart Fail Rev*. 2007;12(2):143-147. doi:10.1007/s10741-007-9017-2 [PubMed 17447137]

37. Hurst S, McMillan M. Innovative solutions in critical care units: extravasation guidelines. *Dimens Crit Care Nurs.* 2004;23(3):125-128. [PubMed 15192356]

38. Ilbawi MN, Idriss FS, DeLeon SY, Berry TE, Duffy CE, Paul MH. Hemodynamic effects of intravenous nitroglycerin in pediatric patients after heart surgery. *Circulation*. 1985;72(3, pt 2):II101-107. [PubMed 3928187]

39. Imani F, Behseresht A, Pourfakhr P, Shariat Moharari R, Etezadi F, Khajavi M. Prevalence of abnormal methemoglobinemia and its determinants in patients receiving nitroglycerin during anesthesia. *Anesth Pain Med*. 2019;9(3):e85852. doi:10.5812/aapm.85852 [PubMed 31497516]

40. "Inactive" ingredients in pharmaceutical products: update (subject review). American Academy of Pediatrics Committee on Drugs. *Pediatrics*. 1997;99(2):268-278. [PubMed 9024461]

41. Jones DW, Ferdinand KC, Taler SJ, et al. 2025 AHA/ACC/AANP/AAPA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PC guideline for the prevention, detection, evaluation and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2025;152(11):e114-e218. doi:10.1161/CIR.0000000000001356 [PubMed 40811497]

42. Kennedy ML, Sowter S, Nguyen H, Lubowski DZ. Glyceryl trinitrate ointment for the treatment of chronic anal fissure: results of a placebo-controlled trial and long-term follow-up. *Dis Colon Rectum*. 1999;42(8):1000-1006. doi:10.1007/BF02236691 [PubMed 10458121]

43. Khera M. Treatment of male sexual dysfunction. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed July 7, 2020.

44. Laitinen P, Happonen JM, Sairanen H, et al. Amrinone Versus Dopamine and Nitroglycerin in Neonates After Arterial Switch Operation for Transposition of the Great Arteries. *J Cardiothorac Vasc Anesth*. 1999;13(2):186-190. [PubMed 10230954]

45. Lee HA, Anderson PO. Giving Partial Doses of Transdermal Patches. *Am J Health Syst Pharm*. 1997;54(15):1759-1760. [PubMed 9262751]

46. London MJ. Intraoperative use of vasoactive agents. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed December 26, 2023.

47. Lowenwirt IP, Zauk RM, Handwerker SM. Safety of intravenous glyceryl trinitrate in management of retained placenta. *Aust N Z J Obstet Gynaecol*. 1997;37(1):20-24. doi:10.1111/j.1479-828x.1997.tb02212.x [PubMed 9075542]

48. Lund JN, Scholefield JH. A randomised, prospective, double-blind, placebo-controlled trial of glyceryl trinitrate ointment in treatment of anal fissure. *Lancet*. 1997;349(9044):11-14. doi:10.1016/S0140-6736(96)06090-4 [PubMed 8988115]

49. Macones G. Puerperal uterine inversion. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed October 12, 2021.

50. Marik PE, Rivera R. Hypertensive emergencies: an update. *Curr Opin Crit Care*. 2011;17(6):569-580. doi:10.1097/MCC.0b013e32834cd31d [PubMed 21986463]

51. Mebazaa A, Tolppanen H, Mueller C, et al. Acute heart failure and cardiogenic shock: a multidisciplinary practical guidance. *Intensive Care Med.* 2016;42(2):147-163. doi:10.1007/s00134-015-4041-5 [PubMed 26370690]

52. Minitran (nitroglycerin) [prescribing information]. Bridgewater, NJ: Valeant Pharmaceuticals North America LLC; December 2014.

53. Nitro-Bid (nitroglycerin) [prescribing information]. Melville, NY: Savage Pharmaceuticals; September 2011.

54. Nitro-Dur (nitroglycerin) [prescribing information]. Orlando, FL: Ingenus Pharmaceuticals LLC; July 2022.

55. Nitro-Dur (nitroglycerin) [product monograph]. Mississauga, Ontario, Canada: Dr. Reddy's Laboratories Canada Inc; March 2019.

56. Nitro-Time (nitroglycerin) [prescribing information]. Farmingdale, NY: Time Cap Labs; November 2025.

57. Nitroglycerin in 5% dextrose injection (nitroglycerin) [prescribing information]. Deerfield, IL: Baxter; August 2016.

58. Nitroglycerin injection [prescribing information]. Shirley, NY: American Regent Inc; November 2005.

59. Nitroject (nitroglycerin) [product monograph]. Montreal, Canada: Juno Pharma Canada Inc; October 2025.

60. Nitrolingual (nitroglycerin) [prescribing information]. Jacksonville, FL: Espero Pharmaceuticals Inc; July 2018.

61. Nitrolingual Pumpspray (nitroglycerin) [prescribing information]. Mason, OH: Independence Pharmaceuticals LLC; May 2024.

62. Nitrolingual Pumpspray (nitroglycerin) [product monograph]. Laval, Quebec, Canada: Sanofi-aventis Canada Inc; June 2021.

63. NitroMist (nitroglycerin) [prescribing information]. Scottsdale, AZ: Evus Pharmaceuticals LLC; June 2022.

64. Nitrostat (nitroglycerin) [prescribing information]. Morgantown, WV: Viatris Specialty LLC; February 2023.

65. Ommen SR, Ho CY, Asif IM, et al. 2024 AHA/ACC/AMSSM/HRS/PACES/SCMR guideline for the management of hypertrophic cardiomyopathy: a report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *Circulation*. 2024;149(23):e1239-e1311. doi:10.1161/CIR.0000000000001250 [PubMed 38718139]

66. O'Sullivan S, Keith MP. Raynaud phenomenon of the Nipple: A Rare Finding in Rheumatology Clinic. *J Clin Rheumatol*. 2011;17(7):371-372. [PubMed 21946463]

67. Park MK, Salamat M. Appendix E: Drugs used in pediatric cardiology. In: *Park's Pediatric Cardiology for Practitioners*. 7th ed. Elsevier Health Sciences; 2021: 477-492.

68. Purdam CR, de Vos RJ. Tendinopathy: Overview of management. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. https://www.uptodate.com. Accessed June 5, 2025.

69. Rangel-Castilla L, Gopinath S, Robertson CS. Management of intracranial hypertension. *Neurol Clin*. 2008;26(2):521-541. [PubMed 18514825]

70. Rao SV, O'Donoghue ML, Ruel M, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI guideline for the management of patients with acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2025;151(13):e771-e862. doi:10.1161/CIR.0000000000001309 [PubMed 40014670]

71. Rectiv (nitroglycerin) [prescribing information]. North Chicago, IL: AbbVie Inc; July 2024.

72. Rectiv (nitroglycerin) intra-anal ointment 0.4% [prescribing information]. North Chicago, IL: AbbVie Inc; April 2024.

73. Reeder GS. Nitrates in the management of acute coronary syndrome. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed May 13, 2025.

74. Refer to manufacturer's labeling.

75. Regitz-Zagrosek V, Roos-Hesselink JW, Bauersachs J, et al; ESC Scientific Document Group. 2018 ESC guidelines for the management of cardiovascular diseases during pregnancy. *Eur Heart J*. 2018;39(34):3165-3241. doi:10.1093/eurheartj/ehy340 [PubMed 30165544]

76. Restieaux M, Maw A, Broadbent R, Jackson P, Barker D, Wheeler B. Neonatal extravasation injury: prevention and management in Australia and New Zealand - a survey of current practice. *BMC Pediatr*. 2013;13:34. [PubMed 23497004]

77. Reynolds PM, MacLaren R, Mueller SW, Fish DN, Kiser TH. Management of extravasation injuries: a focused evaluation of noncytotoxic medications. *Pharmacotherapy*. 2014;34(6):617-632. [PubMed 24420913]

78. RHO-Nitro Pumpspray (nitroglycerin sublingual spray) [product monograph]. Boucherville, Quebec, Canada: Sandoz Canada Inc; June 2022.

79. Roche Gamón E, De la Cuadra Oyanguren J, Pérez Ferriols A, Fortea Baixauli JM. Contact dermatitis from nitroglycerin in a transdermal therapeutic system. *Acta Derm Venereol*. 2006;86(6):544-545. doi:10.2340/00015555-0146 [PubMed 17106605]

80. Sawatzky-Dickson D, Bodnaryk K. Neonatal intravenous extravasation injuries: evaluation of a wound care protocol. *Neonatal Netw*. 2006;25(1):13-19. [PubMed 16514862]

81. Shehab N, Lewis CL, Streetman DD, Donn SM. Exposure to the pharmaceutical excipients benzyl alcohol and propylene glycol among critically ill neonates. *Pediatr Crit Care Med*. 2009;10(2):256-259. [PubMed 19188870]

82. Smith ZR, Horng M, Rech MA. Medication-induced hyperlactatemia and lactic acidosis: a systematic review of the literature. *Pharmacotherapy*. 2019;39(9):946-963. [PubMed 31361914]

83. Stefanos SS, Kiser TH, MacLaren R, Mueller SW, Reynolds PM. Management of noncytotoxic extravasation injuries: a focused update on medications, treatment strategies, and peripheral administration of vasopressors and hypertonic saline. *Pharmacotherapy.* 2023;43(4):321-337. doi:10.1002/phar.2794 [PubMed 36938775]

84. Stendra (avanafil) [prescribing information]. Chesterbrook, PA: Auxilium; April 2014.

85. Taylor T, Kennedy D. Safety of topical glyceryl trinitrate in the treatment of anal fissure in breastfeeding women. *Birth Defects Res A Clin Mol Teratol*. 2008;82:411.

86. Thigpen JL. Peripheral intravenous extravasation: nursing procedure for initial treatment. *Neonatal Netw*. 2007;26(6):379-384. [PubMed 18069429]

87. Thomas J, Saenz E, Haggerty P. Nitroglycerin-induced asystole: a common therapy with a rare adverse effect. *Am J Med*. 2016;129(10):e245-e246. doi:10.1016/j.amjmed.2016.06.039 [PubMed 27430406]

88. Tobias JD, Naguib A, Simsic J, Krawczeski CD. Pharmacologic control of blood pressure in infants and children. *Pediatr Cardiol*. 2020;41(7):1301-1318. doi:10.1007/s00246-020-02448-2 [PubMed 32915293]

89. Treadwell T. The management of intravenous infiltration injuries in infants and children. *Ostomy Wound Manage*. 2012;58(7):40-44. [PubMed 22798353]

90. Wagenknecht DM, Baaske DM, Alam AS, Carter JE, Shah J. Stability of nitroglycerin solutions in polyolefin and glass containers. *Am J Hosp Pharm*. 1984;41(9):1807-1811. [PubMed 6437221]

91. Weeks A. Retained placenta after vaginal birth. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed September 22, 2020.

92. Williams RS, Mickell JJ, Young ES, et al. Methemoglobin Levels During Prolonged Combined Nitroglycerin and Sodium Nitroprusside Infusions in Infants After Cardiac Surgery. *J Cardiothorac Vasc Anesth*. 1994;8(6):658-662. [PubMed 7880995]

93. Wong AF, McCulloch LM, Sola A. Treatment of Peripheral Tissue Ischemia With Topical Nitroglycerin Ointment in Neonates. *J Pediatr*. 1992;121(6):980-983. [PubMed 1447671]

94. Zamani M, Alimi R, Arabi SM, Moradi M, Azmoude E. Comparison of the efficacy of nifedipine with ritodrine, nitroglycerine and magnesium sulfate for the management of preterm labor: a systematic review and meta-analysis. *BMC Pregnancy Childbirth*. 2024;24(1):318. doi:10.1186/s12884-024-06497-w [PubMed 38664622]

95. Zar T, Graeber C, Perazella MA. Recognition, treatment, and prevention of propylene glycol toxicity. *Semin Dial*. 2007;20(3):217-219. [PubMed 17555487]

96. Zuberi BF, Rajput MR, Abro H, Shaikh SA. A randomized trial of glyceryl trinitrate ointment and nitroglycerin patch in healing of anal fissures. *Int J Colorectal Dis*. 2000;15(4):243-245. doi:10.1007/s003840000230 [PubMed 11008725]

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